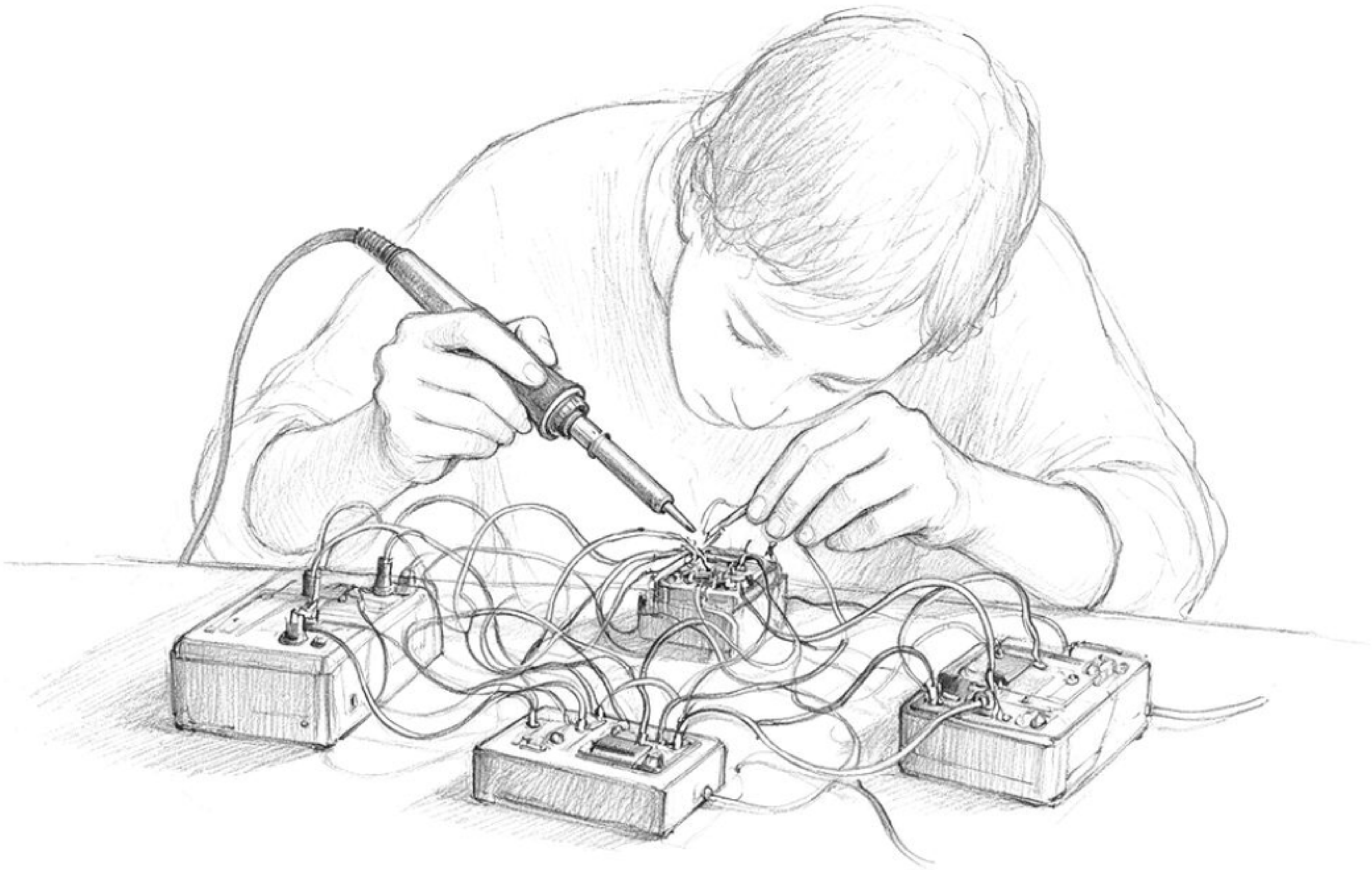
MODULE 13 · AI & ML Foundations

The Agent Ecosystem: MCP, Tools & Agentic IDEs

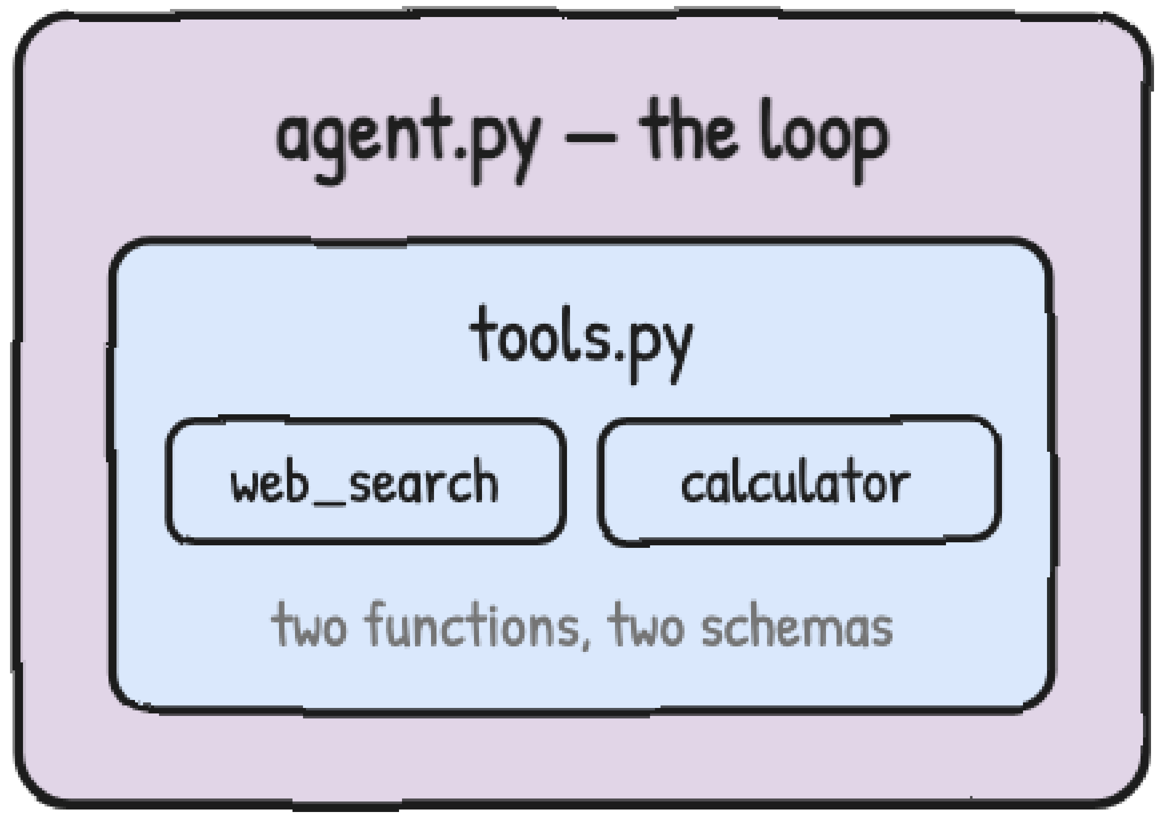
You built an agent by hand. Now name every part of it.

Gourav Shah · School of DevOps & AI · Lesson + Lab + Quiz



The problem with hand-wiring every tool

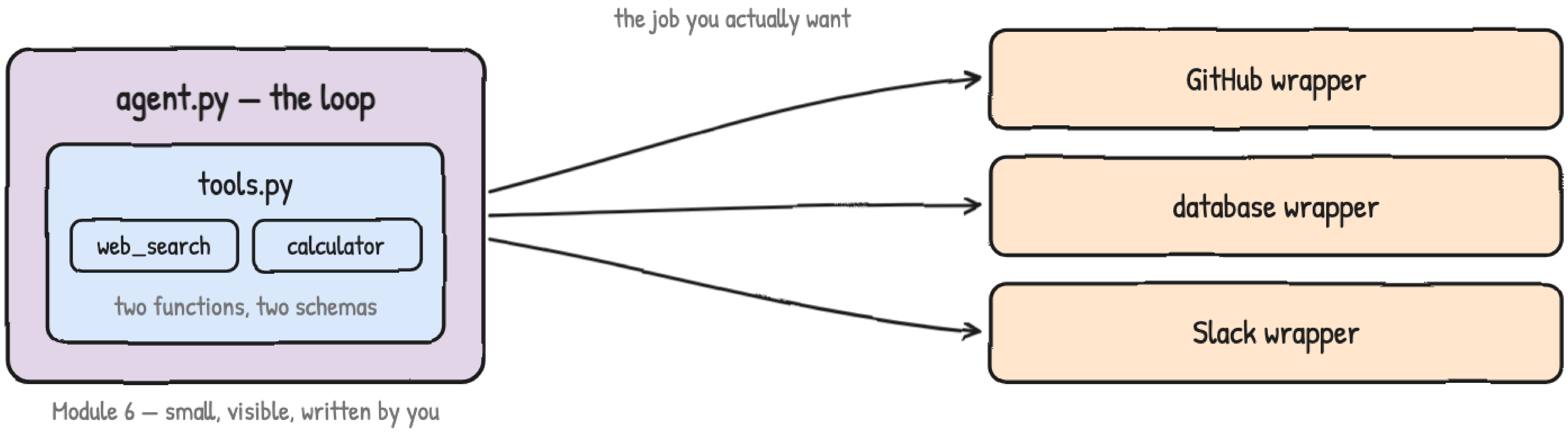
Every tool wired straight into one script



Module 6 — small, visible, written by you

It worked. It just doesn't travel.

Every tool wired straight into one script



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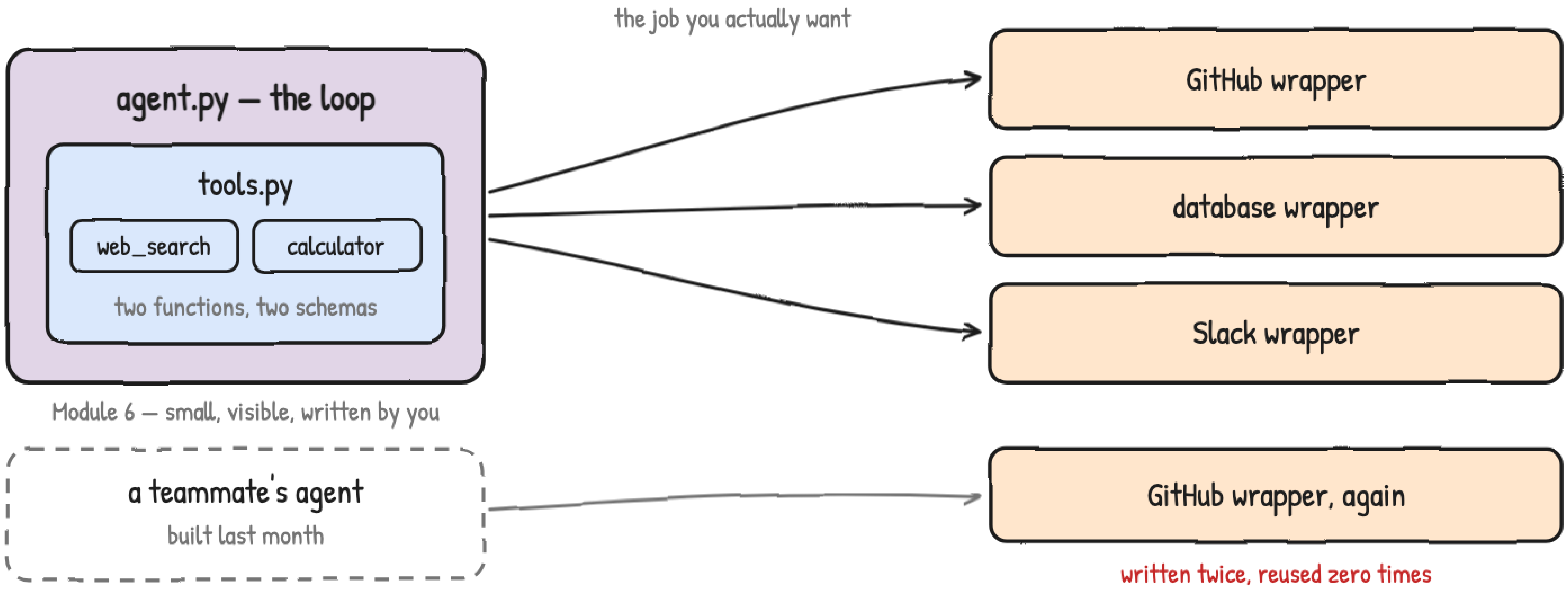

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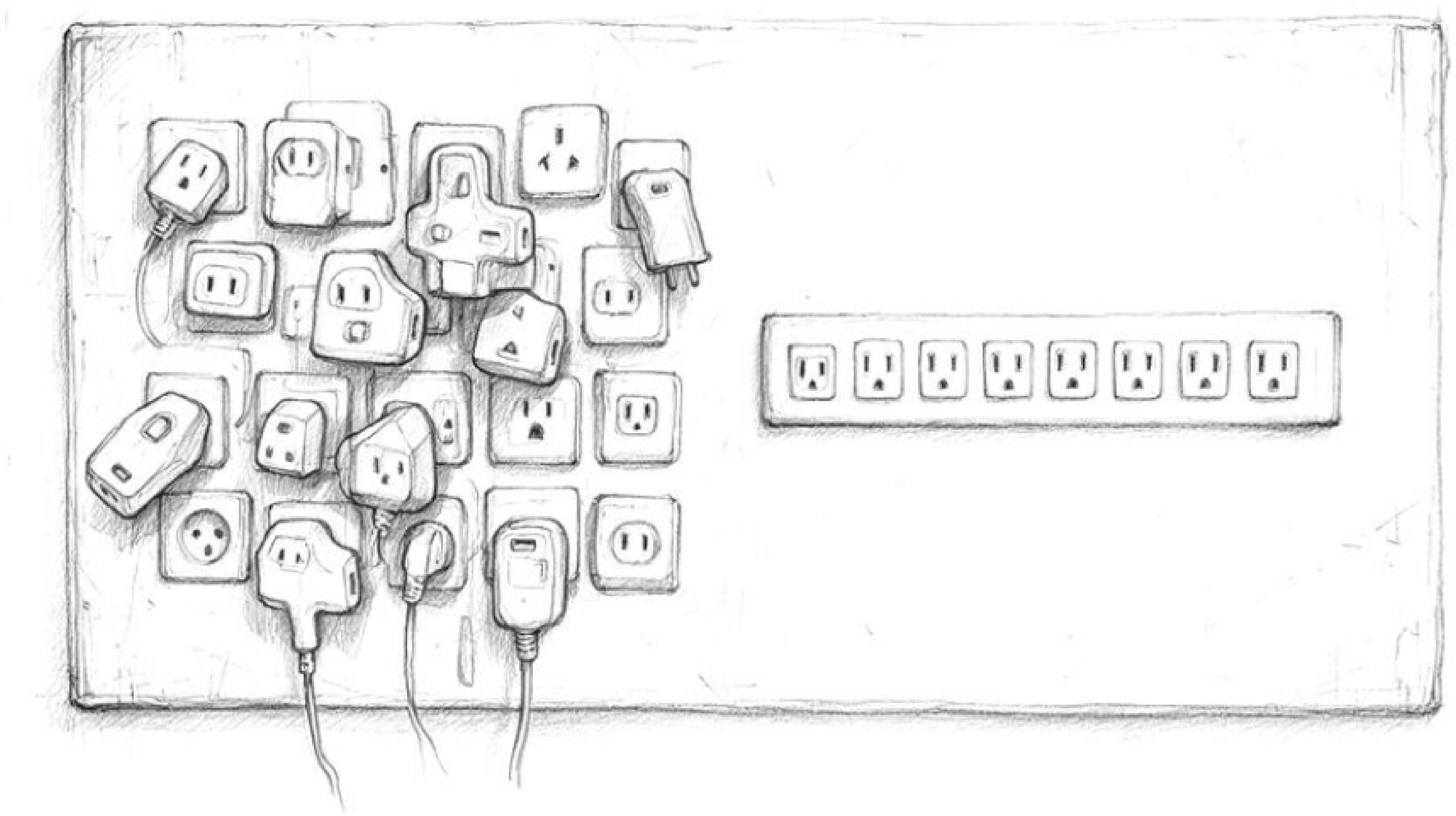
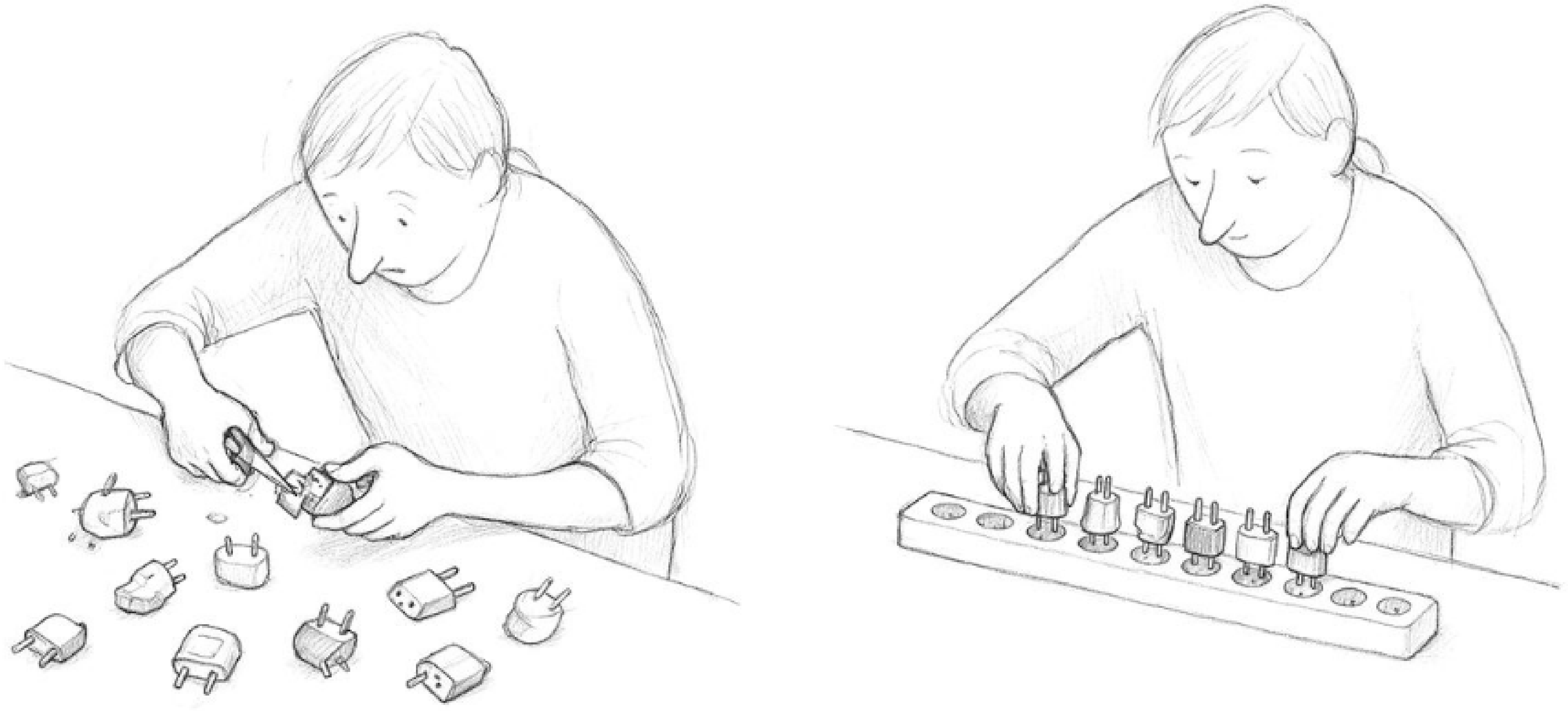


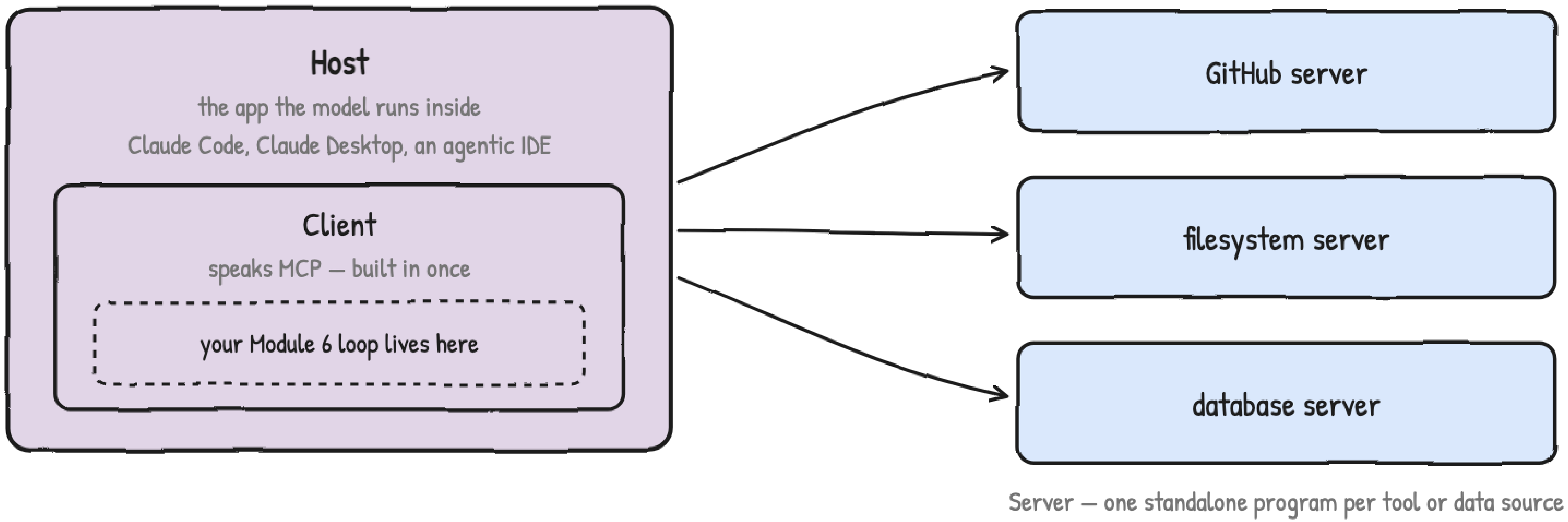
MCP: a standard, not a specific tool

Before a shared standard, every pairing needed its own code



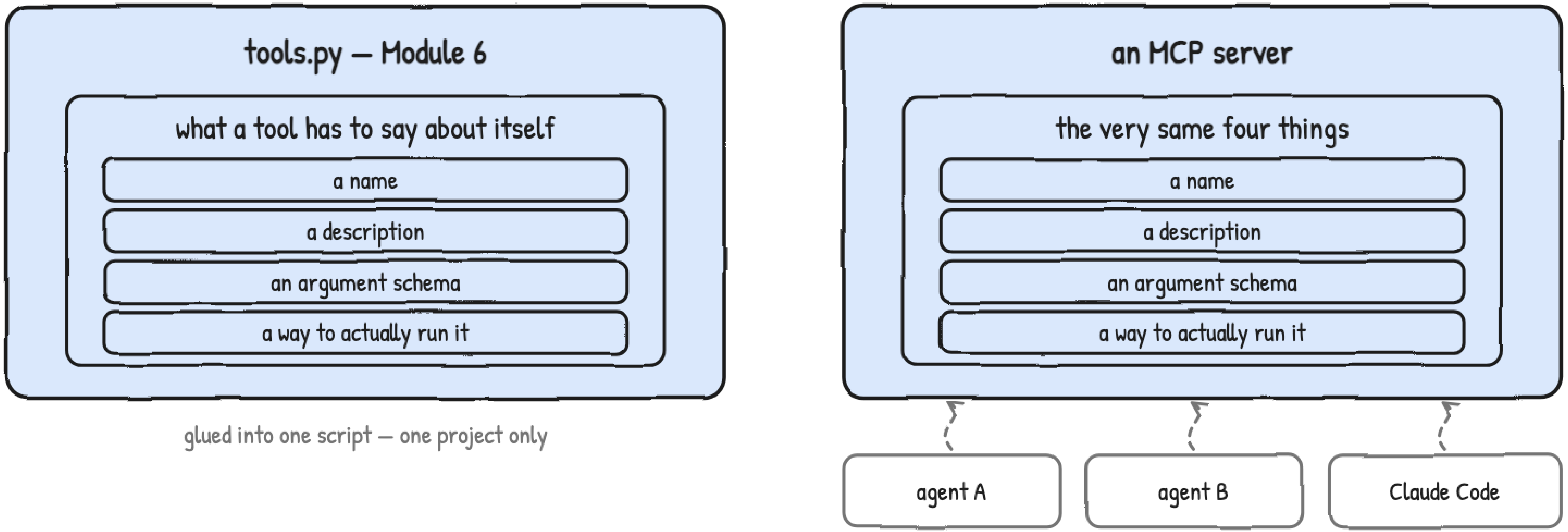
Agree on one shape, and the extra work goes away.

MCP names three roles: host, client, server

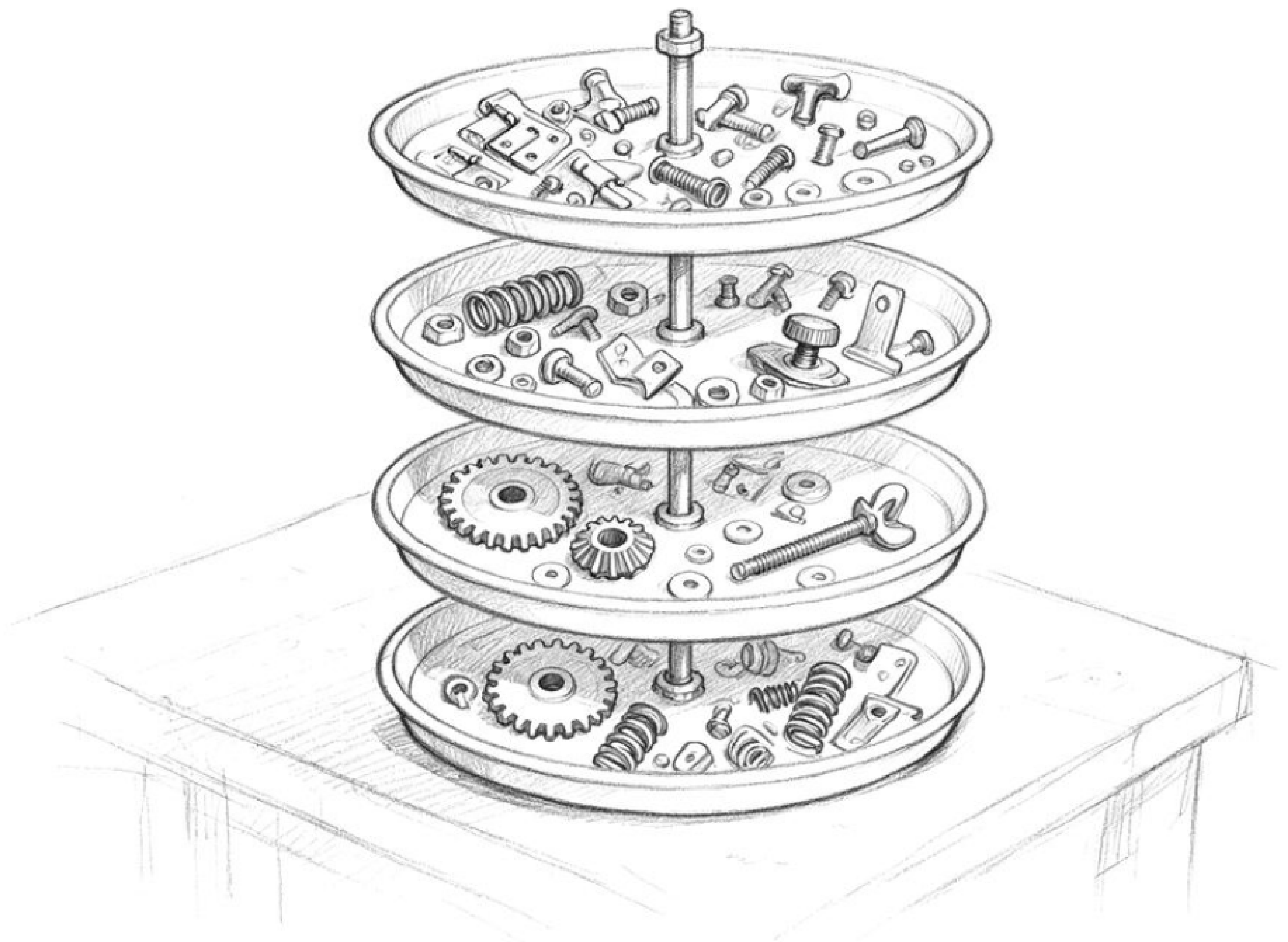


Three words that let you read any MCP setup.

Same job as tools.py, but standalone and shared



Write it once. Every MCP host can call it.



The agent stack, layer by layer

Every agent is built from four layers

Model, tools, memory, orchestration.

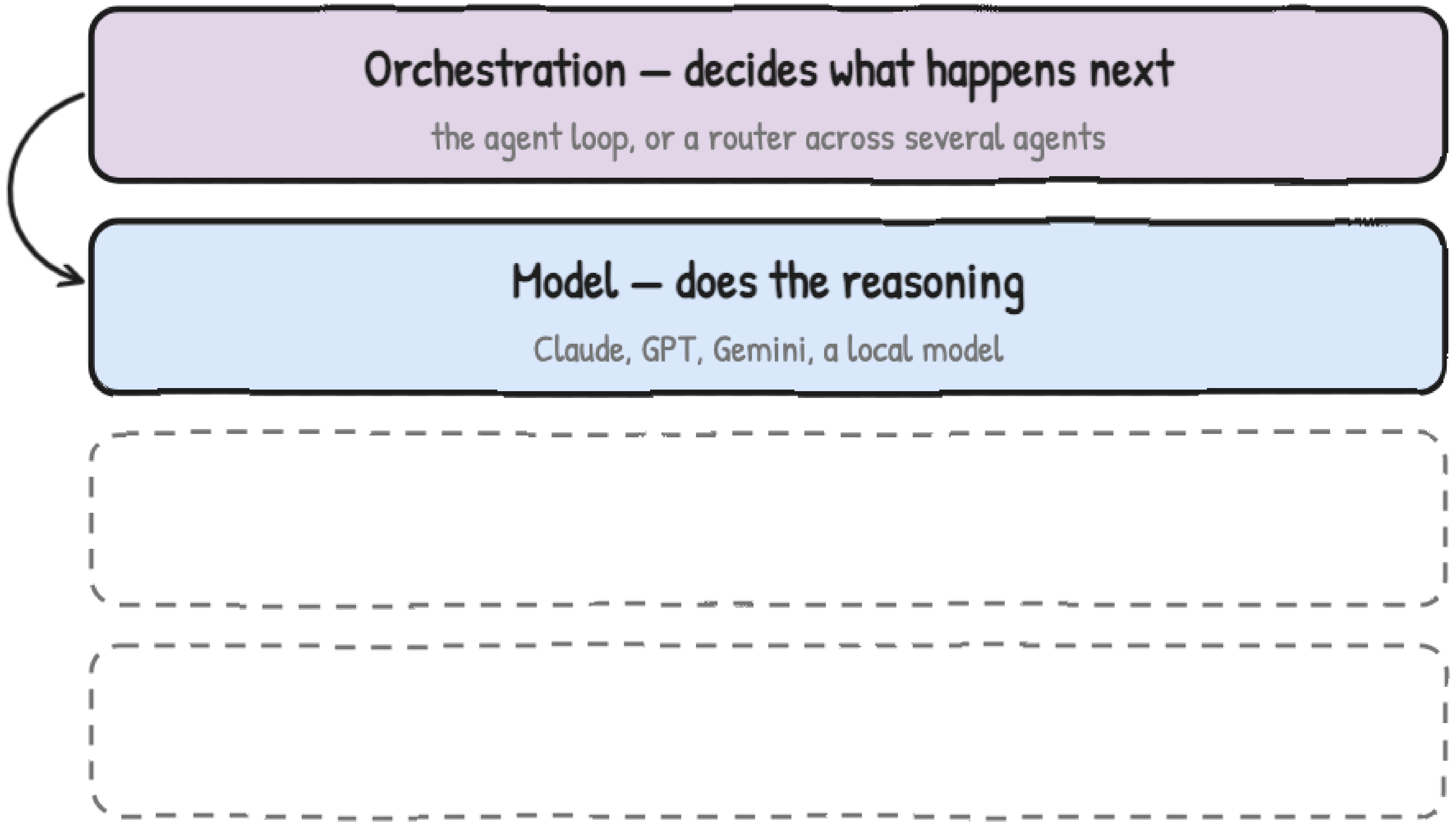
Every agent is built from four layers

Orchestration — decides what happens next

the agent loop, or a router across several agents

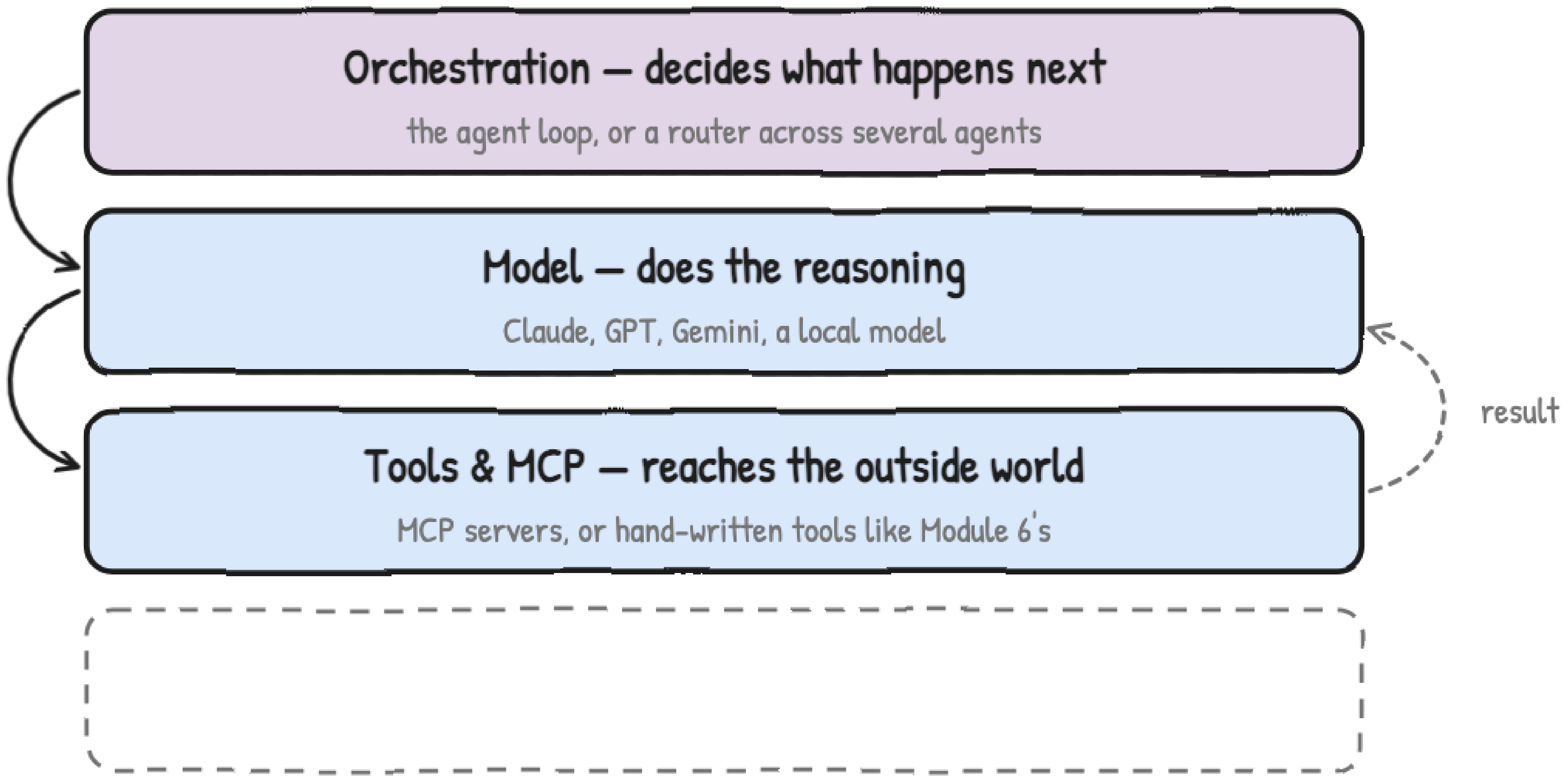
Model, tools, memory, orchestration.

Every agent is built from four layers



Model, tools, memory, orchestration.

Every agent is built from four layers



Model, tools, memory, orchestration.

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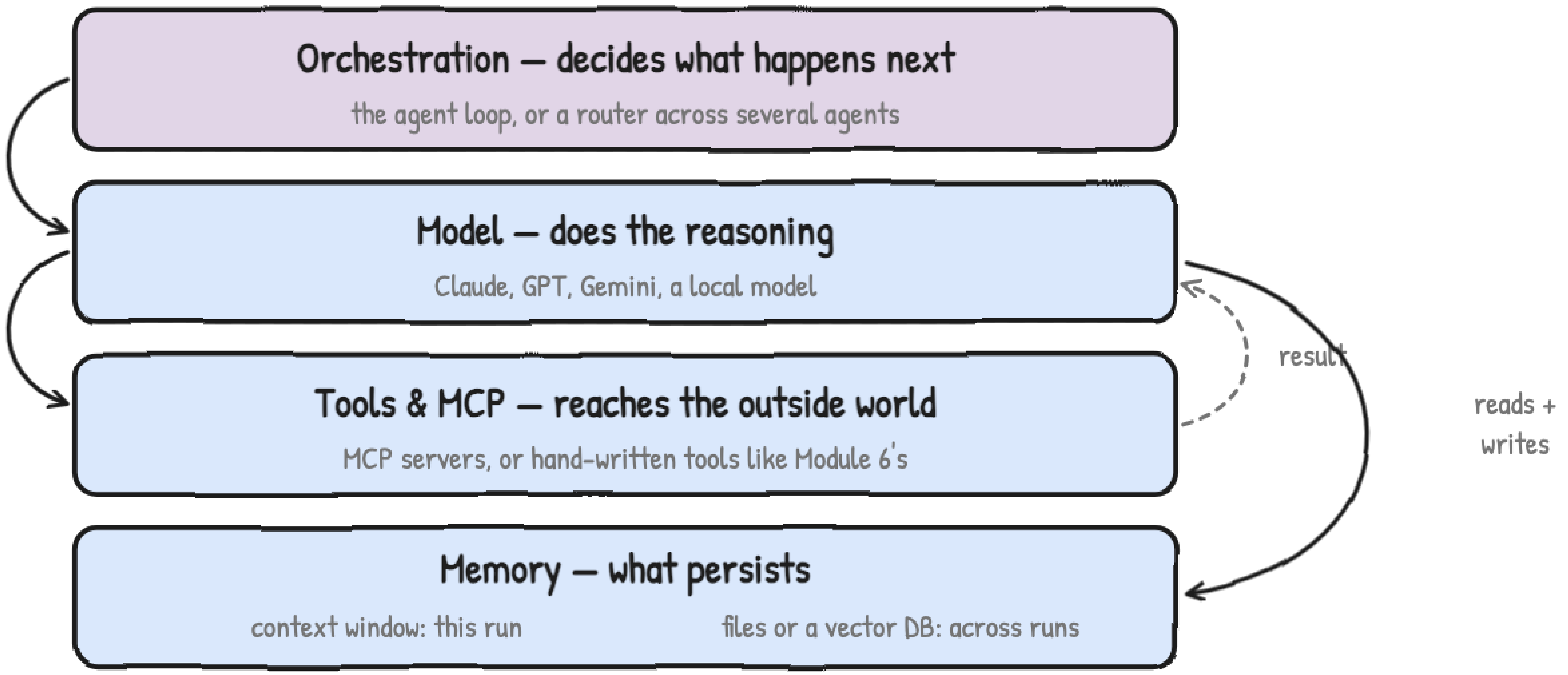


Model, tools, memory, orchestration.

Module 6 already had all four layers

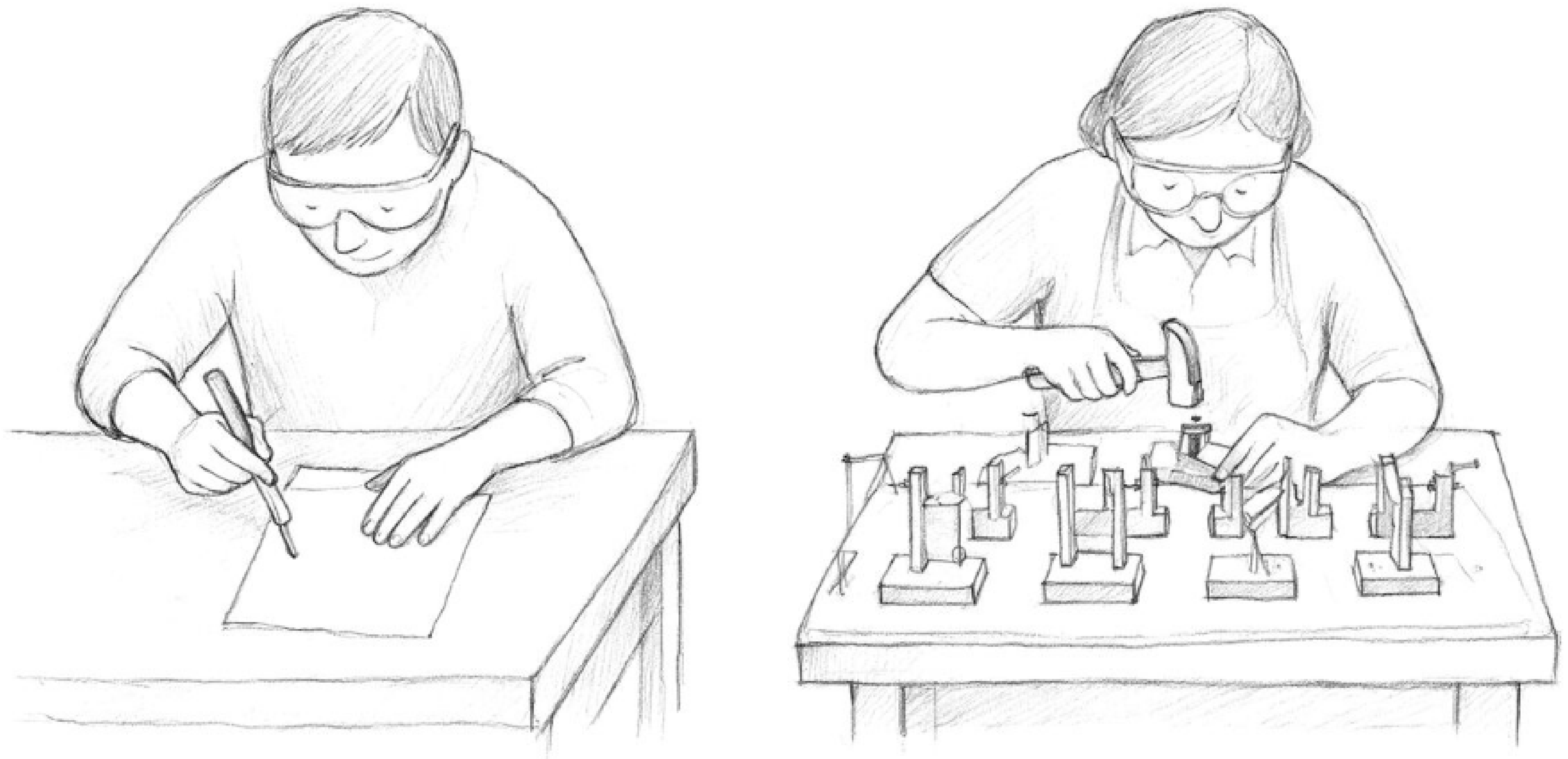
1. **Orchestration**
   the loop in agent.py, stopped by MAX_ITERATIONS
2. **Model**
   whatever you set in .env — Gemini, or a local Ollama model
3. **Tools & MCP**
   tools.py — hand-written, doing the identical job
4. **Memory**
   the running conversation, plus Lab 5's Chroma store for the long-term half

nothing about this shape changes as systems get bigger

You built them. You just had no names for them.

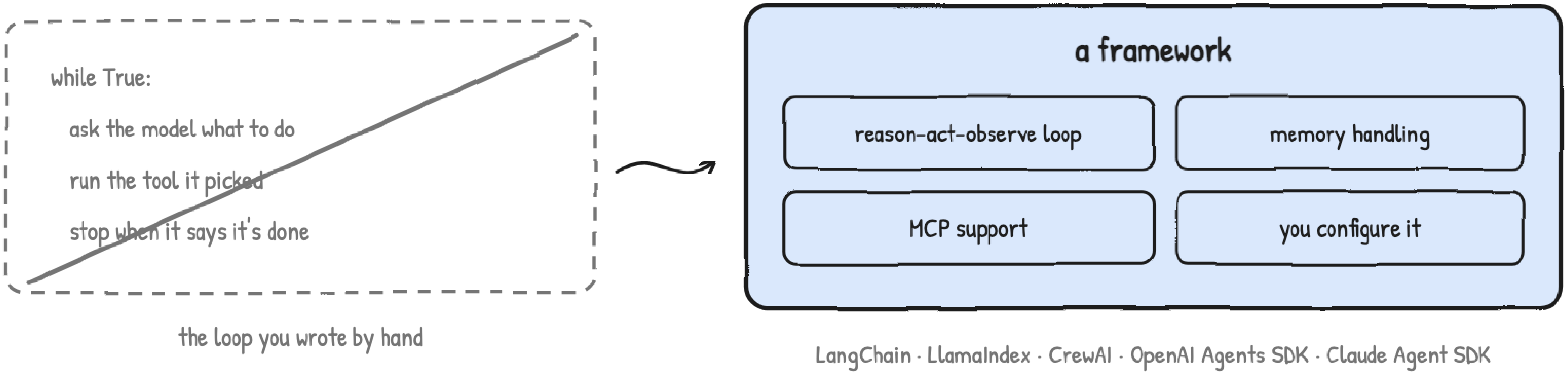
The agent tooling landscape

Same work — the question is how much is built in



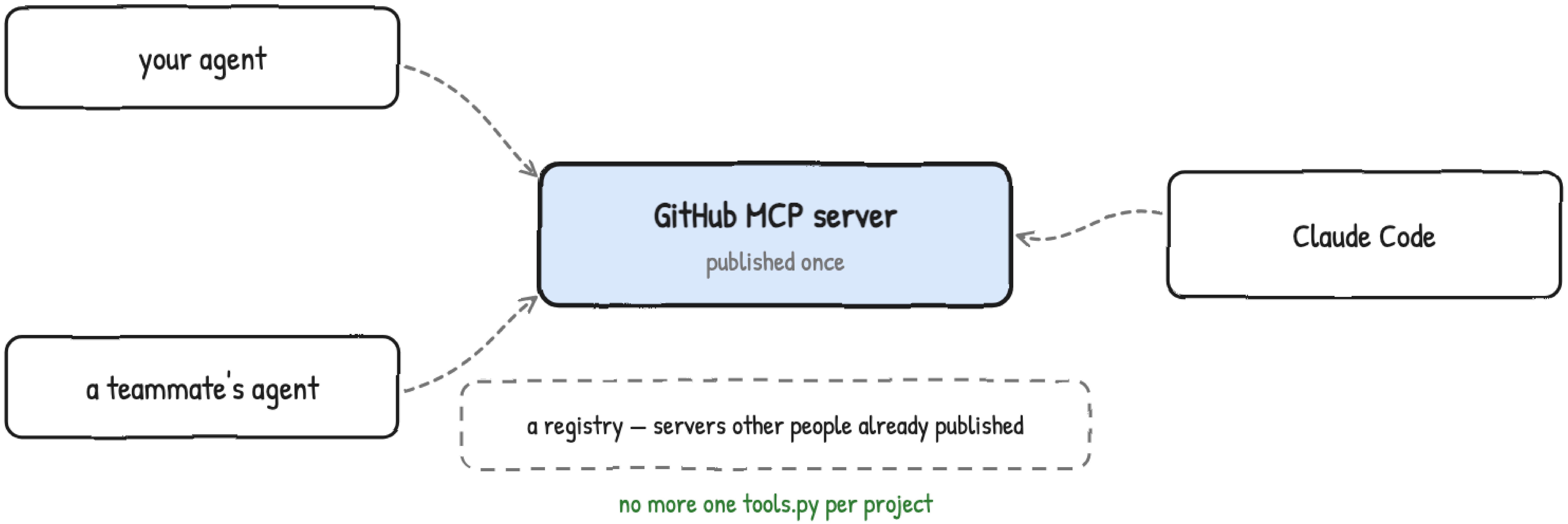
A bare desk, or a bench with the jigs already fitted.

A framework packages the loop you wrote by hand



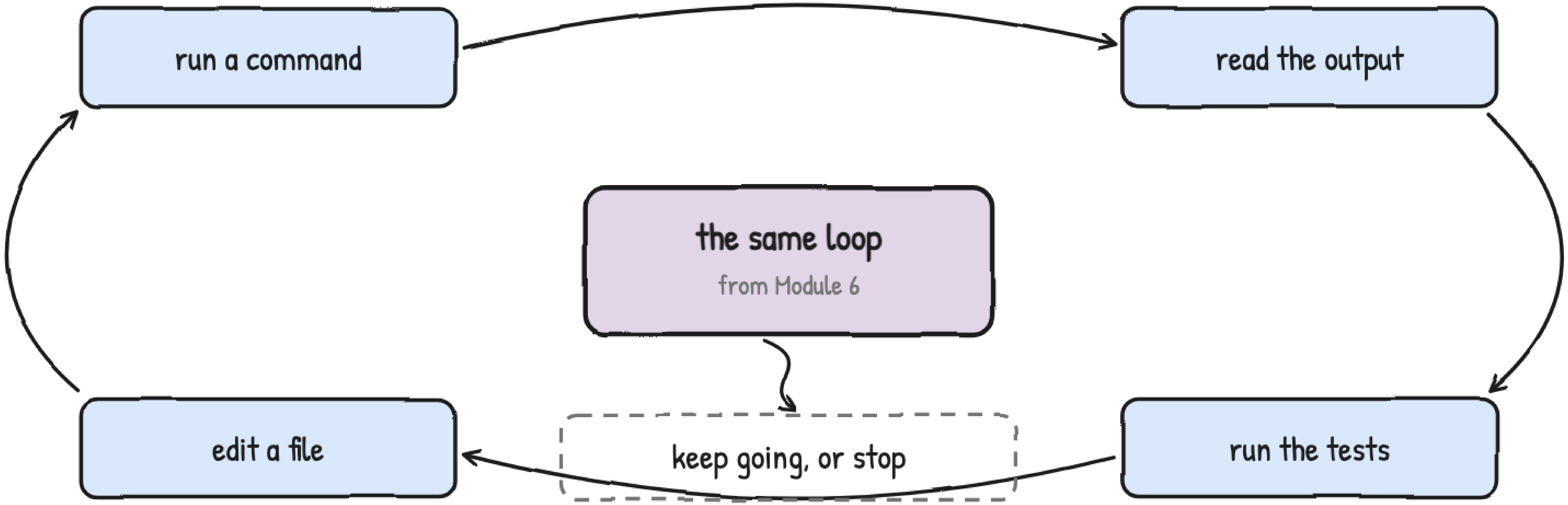
You configure the loop instead of writing it.

Write the connector once, reuse it everywhere



One server, every host that speaks MCP.

An agentic IDE runs that same loop over your codebase



Claude Code · Cursor · Windsurf · Copilot agent mode

Scoped to your codebase, not a search engine.

Multi-agent patterns, at a foundational level

One agent is still the normal answer

one model, one toolset, one loop

start to finish, every time

most of what you build

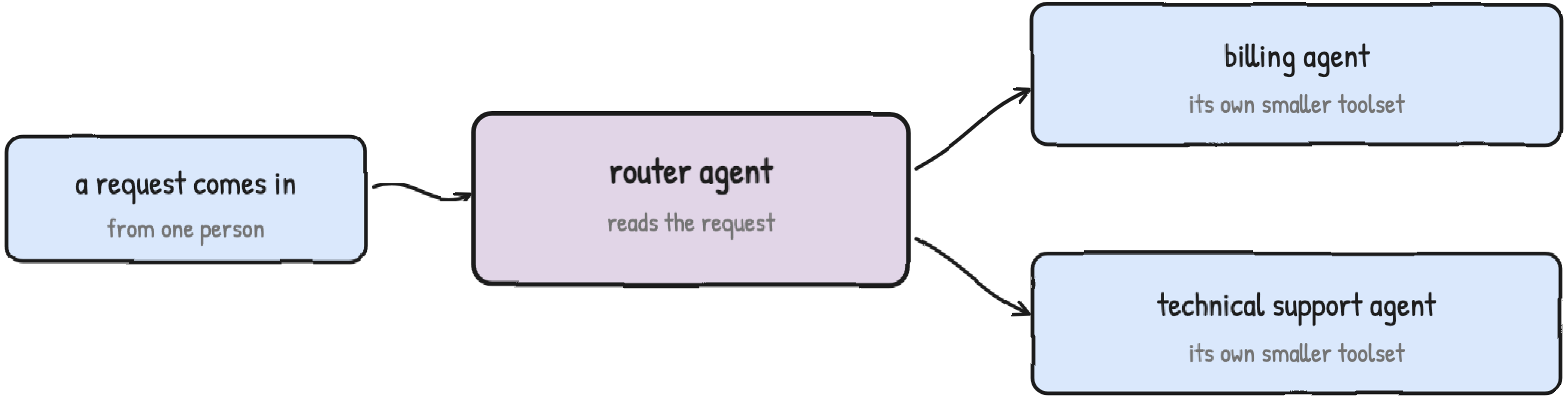
when the task outgrows one loop

router | pipeline | fan-out

three patterns, next

Split the work only when one loop can't hold it.

A router reads the request and hands it over



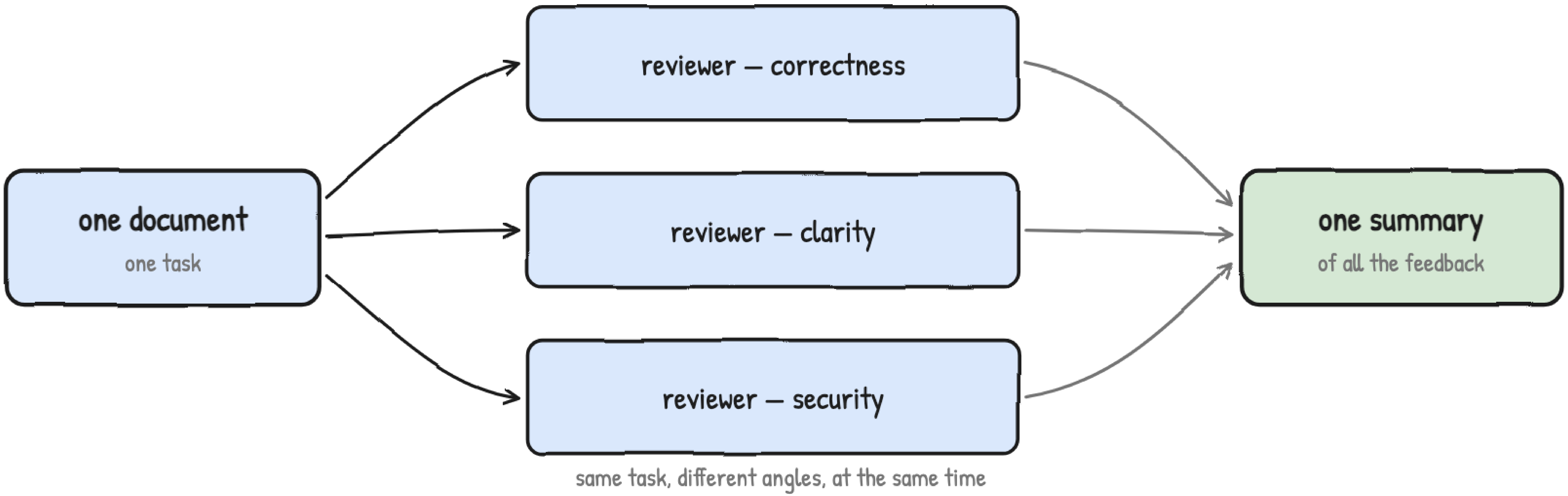
Module 6's triage desk, one level up

Orchestrator plus specialists.

One agent's output is the next one's input

A sequential pipeline, in a fixed order.

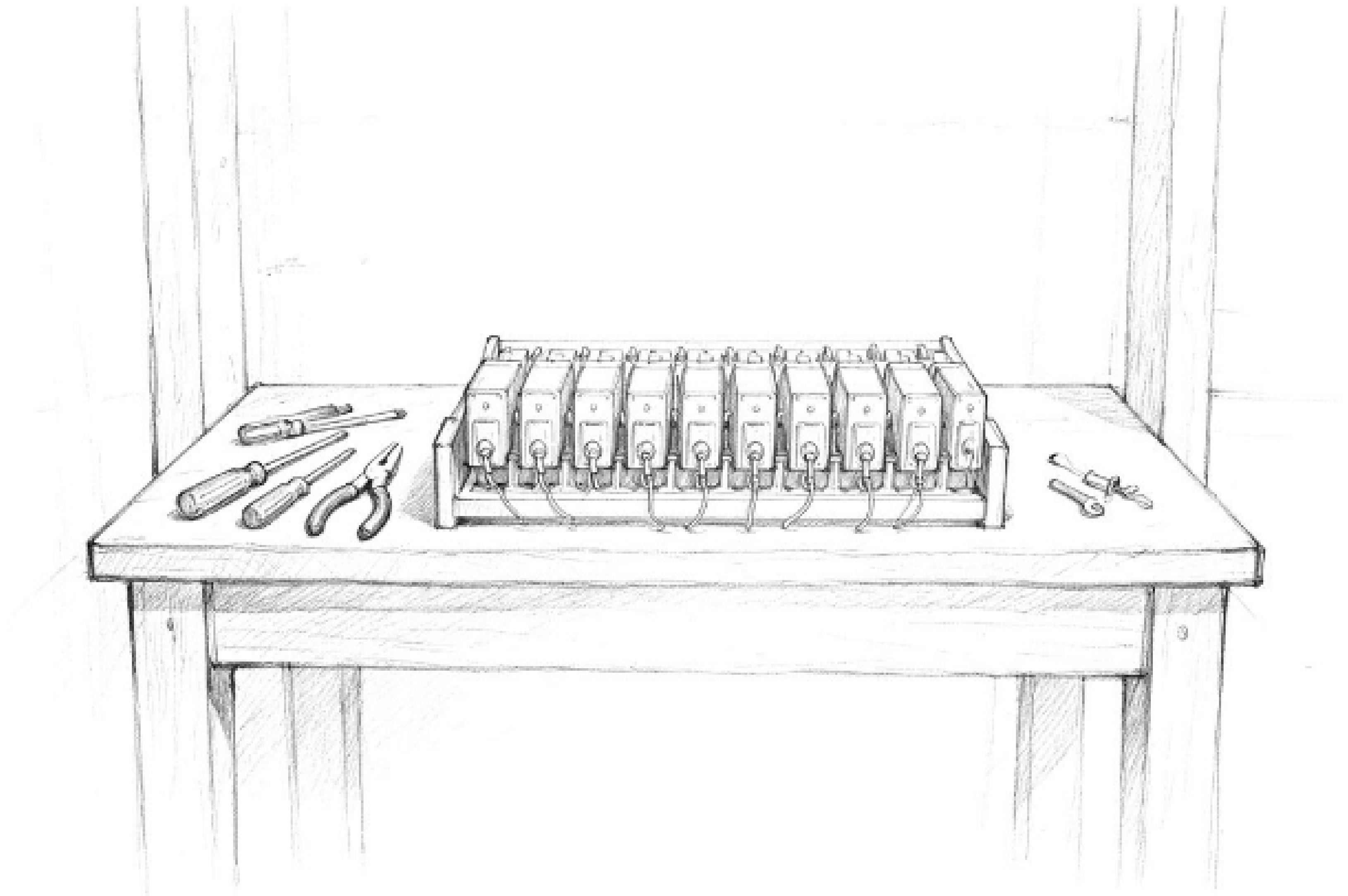
Many angles at once, then one summary



Fan out to several agents, fan back in to one.

MODULE 13 TAKEAWAYS

You can now name every layer under any agent product



1. Orchestration — the loop, or a router
2. Model — does the reasoning
3. Tools & MCP — reaches the outside world
4. Memory — this run, and across runs

same loop underneath — frameworks and IDEs just build more of it for you

Now open **Module 13 · Lab.** · **Gourav Shah** · School of DevOps & AI